

Hugo Thomas

Race Engineer | Telemetry & Data Analysis | Simulation Software | Python Automation

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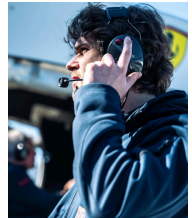
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NAT French



MY LIFE PHILOSOPHY

"I'm pulled toward challenges." *Hard ones, where the conditions are tough, and the answer isn't obvious. That's the thread through everything I've done - batteries, GT3, LMDh, race engineering, simulation software. The field kept changing, but the motivation never did.*

SUMMARY

Motorsport engineer (Master's in Mechanical Design, HEI Lille) with **2 years spanning race engineering, telemetry analysis and simulation software development**. **Race Engineer at Octane126** in the **Ferrari Challenge Trofeo Pirelli Pro**: managed a **6-person race team**, led on-track setup and strategy, and took **4 podiums including a race win** with car 26. Built **Python telemetry automation** - an automated stint report generating KPIs twice daily from raw **MoTeC and WinTAX** exports, saving 1-2 hours per report - and led a **6-developer software team** building a full racing simulation platform from a concept and **Figma** prototype I created. Previously contributed to the **Ferrari 296 GT3** and **Alpine LMDh A424** programmes at **ORECA** as Assembly Method Engineer. Analytical, hands-on, effective in fast-paced cross-functional environments. French native, **English C1**; Zürich, Swiss Permit B, available immediately and free to travel to events.

EXPERIENCE

Race Engineer · Project Manager / Lead Engineer, Simulation Software
Octane126 · 4 podiums incl. 1 win · car 26

02/2024 - 12/2025
Zürich, Switzerland

- Led **on-track setup and strategy**, customising and applying a **setup Wizard tool** tailored to car performance, track conditions and driver skill.
- Collaborated on **car model development**: suspension pick-up points, gearbox ratios, engine curves, mass and motion ratios.
- Managed a 6-person race team** - coordinating roles, assigning tasks and driving collaboration - including event logistics: truck readiness and cross-border authorisations.
- Achieved **4 podium finishes in the season, including one win**, in the Ferrari Challenge Trofeo Pirelli Pro with car 26.
- Developed an automated **Python stint report** generating key KPIs and comments **twice daily from raw MoTeC and WinTAX exports** - saving **1-2 hours per report** and enabling focused driver communication and car tuning.
- Created the concept and designed the **UI prototype in Figma** for a complete racing-team software solution: **lap-time simulation, run plans, data analysis and parts-mileage tracking**.
- Managed an external development team of 6 developers and 2 project managers** through daily meetings and report reviews - planning, budgeting, handover and quality checks.

Assembly Method Engineer (Intern · Ferrari 296 GT3 / Alpine LMDh A424)
Oreca · 6 Months

03/2023 - 08/2023
Signes, FR

- Supported the development and optimisation of **assembly processes and documentation** for the build of the **Ferrari 296 GT3** and **Alpine LMDh A424** race cars.
- Improved collaboration between **workshop and design office** by initiating and leading **weekly coordination meetings**, streamlining communication and workflows.
- Learned how a top-level race organisation keeps a moving car definition buildable - the practical side that later made me a better race engineer.

Data Engineer (Intern · GT3 customer racing)
Sainteloc · 4 Months

11/2022 - 02/2023
Saint-Étienne, FR

- Developed **Excel VBA data-analysis tools** that automated and streamlined **on-track performance-metrics reporting**, improving decision speed and accuracy during races.
- Designed, procured parts for and assembled a **spring-rate measurement bench** (SolidWorks and welding), enabling **continuous monitoring of spring-rate evolution across the racing season** to support setup and reliability.

Mechanical Design Engineer (Intern · Rail)
Socofer · 3 Months

05/2022 - 07/2022
Tours, FR

- Conducted **daily traction, braking and acceleration calculations** for railway vehicles against safety and performance standards - vehicle-performance maths applied every day.
- Designed and modelled a **lithium-ion battery housing with an integrated water-pipeline cooling system** (SolidEdge) to prevent thermal-runaway events.

SKILLS

Race engineering	On-track setup & strategy (Ferrari Challenge) · Setup Wizard tool customisation & application · Run plans · Driver communication & car tuning · Race-team management (6 people) · Event logistics & cross-border authorisations
Data & telemetry	MoTeC i2 Pro · WinTAX · Sysma · Telemetry & acquisition analysis · KPI definition & stint reporting · Performance-metrics reporting · Data-driven setup decisions
Simulation	ChassisSim · Lap-time simulation (platform concept & specification) · Car model development - suspension pick-up points, motion ratios, gearbox ratios, engine curves, mass · FEA - Abaqus (static & modal)
Software & automation	Python - telemetry automation, KPI reporting pipelines · Excel VBA - data-analysis tooling · HTML / CSS · Figma - UI concept & prototyping · Building tools race engineers actually use
Leadership & delivery	Led a 6-developer / 2-PM external team - planning, budgeting, handover, quality checks · Team leadership & task allocation · Project management · Cross-functional communication
Engineering & CAD	SolidWorks · SolidEdge · Test-bench design, build & instrumentation · Vehicle performance calculation (traction, braking, acceleration) · Mechanical & hydraulic transmission

PROJECTS

The Finite Element Method for Linear Structural Analysis (Udemy)	01/2026
• Structured FEA training in Abaqus: linear static and modal analysis, meshing, boundary conditions, load cases and critical assessment of results.	
Mechanical Design Project - Gravitational Feed Conveyor	01/2023
• Team mechanical design project (240 hours) on a gravitational feed conveyor - concept through detailed design.	
Data Driven Performance Engineering Seminar (MoTeC acquisition & telemetry)	03/2023
• Applied advanced vehicle-dynamics concepts to race cars through hands-on analysis with the MoTeC data acquisition and telemetry tool.	
• Deepened understanding of vehicle behaviour and performance optimisation.	
Team Leader - Unizik Music Association	2023
• Led a 23-member association; organised jams and festivals - logistics, people and deadlines.	

LANGUAGES

French	Native	English	C1	German	B1	Spanish	B1
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EDUCATION

Master in Mechanical Design	01/2018 - 01/2024
HEI Lille	Lille, France
• Mechanical Design · Structural Design (Finite Element Method) · Strength of Materials	
• Mechanical and Hydraulic Transmission	
Scientific Baccalaureate	06/2018
Lycée Notre Dame la Riche	Tours, France
• Specialised in Physics, with Honours (14.02 / 20).	

INTERESTS

- **Motorsport technology research** - technical advancements and latest innovations
- **Physical activities** - hiking, climbing, gym, tennis, golf
- **Guitarist and composer** - self-taught in music production
- **Traveling** - one-month solo road trip in Italy; trip to Thailand